

MASS-ENG-007

Knowledge Engine Specification

Page 1 – Purpose & Scope

Purpose

Define the enterprise knowledge subsystem responsible for storing, organizing, indexing, governing, and retrieving institutional knowledge across MASS HQ.

Objectives

- Provide a centralized knowledge repository
- Support structured and unstructured content
- Enable semantic search and retrieval
- Preserve enterprise knowledge over time

Out of Scope

Document rendering, workflow execution, and AI reasoning.

Page 2 – Architecture

Core Components

Knowledge Repository, Metadata Catalog, Search Index, Taxonomy Manager, Knowledge API, Version Manager.

Relationships

Consumes Enterprise Core, Identity, Security, and Event Bus. Serves Workflow, AI, Documents, and all business modules.

Page 3 – Functional Specification

Requirements

1. Store knowledge assets.
2. Maintain metadata and taxonomy.
3. Support version history.
4. Provide search and retrieval APIs.
5. Publish knowledge lifecycle events.

Non-Functional

Fast retrieval, immutable history, extensible metadata, auditability, scalability.

Page 4 – Interfaces & Examples

Interfaces

CreateKnowledge(), UpdateKnowledge(), SearchKnowledge(), RetrieveKnowledge(), ArchiveKnowledge().

Example Flow

Policy Published → Knowledge Engine indexes content → Event published → AI Engine refreshes context → Users discover updated knowledge through search.

Page 5 – Construction Package

Claude Checklist

Implement repository; metadata catalog; taxonomy service; indexing; search API; versioning; event integration; authorization checks; unit and integration tests.

Definition of Done

Knowledge assets are securely stored, versioned, searchable, and available through standardized

interfaces.

Constitution References

Knowledge, Governance, Security, Platform Principles, Enterprise Structure.